TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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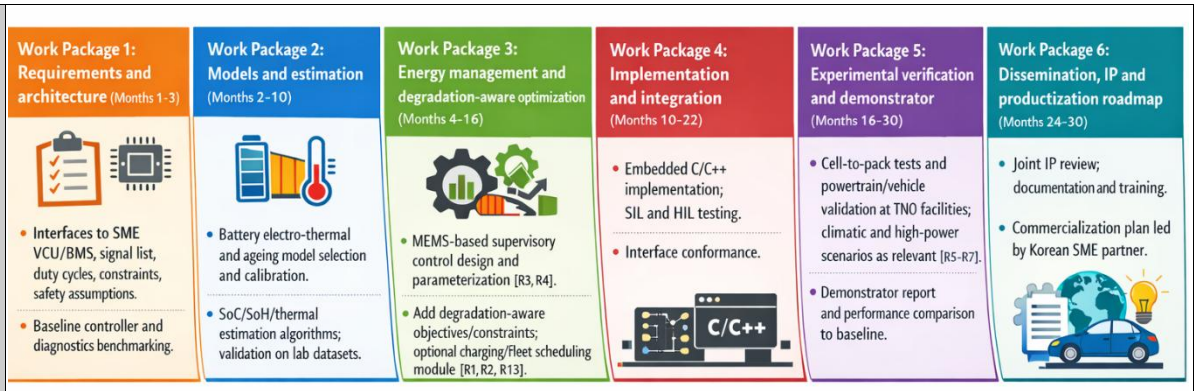
2. Technology Overview

Title of Proposed Technology	Real-time Battery Digital Twin and Hardware-in-the-Loop (HIL) Validation Suite for EV Powertrain Integration and Battery Safety Monitoring			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Machinery & Materials		Information & Communications	
	Medium Category	Small Category	Medium Category	Small Category
	Automotive & Railway Vehicles	System integration technology	Software	Embedded SW
Concept of Proposed Technology	Core concept: a modular, real-time battery digital twin that can be executed in hardware-in-the-loop (HIL) environments to validate EV powertrain control, charging strategies, and battery safety functions, using TNO's Battery Lab expertise and facilities [1]. Main functions:			

	1) Multi-fidelity battery models (equivalent-circuit plus electro-thermal and ageing extensions) suitable for real-time execution and calibration using lab test data. TNO has publicly described its portfolio of testing methods, models, and algorithms within the Battery Lab context [1]. 2) Battery diagnostics and monitoring algorithms, including impedance-based and data-driven methods, building on published work that compares active versus passive monitoring approaches and discusses implementation in automotive contexts [7]. 3) HIL integration layer to connect the battery digital twin to vehicle controllers (BMS, VCU, charger controller) and to perform automated scenario testing, including fault injection. TNO explicitly positions HIL setups as a means to accelerate development in the Battery Lab [1]. 4) Test-data driven parameterisation and validation using high-power cycling and climatic testing capabilities (e.g., pack charge/discharge up to 750 V / 400 A / 300 kW and wide temperature ranges) as reported in TNO technical publications [3][4]. Potential uses: accelerated development and verification of battery management and vehicle control software; validation of charging protocols; and generation of standardised indicators for battery performance reporting and battery passport related metrics (e.g., efficiency and efficiency-fade) [8].
Rationale for Joint R&D Project	EV validation and control-system providers increasingly need validated battery behaviour and safety margins under diverse thermal, ageing, and fast-charging conditions, yet full-vehicle and full-pack testing is expensive and slow. This joint R&D combines: - TNO: battery modelling, diagnostics algorithms, and controlled testing in the Battery Lab and the Innovation Centre for Sustainable Powertrains (ICSP) [1][2]. - Korean SME partner: industrial HIL platforms and EV powertrain integration know-how to deploy the digital twin and diagnostics in development workflows. The collaboration enables faster iteration loops (model-in-the-loop to HIL to lab validation), reduces development risk, and creates a transferable validation product. Relevant building blocks are supported by peer-reviewed and conference publications on EV battery modelling, scheduling and diagnostics [5][6][7].
Key Words	1. Battery digital twin 2. Hardware-in-the-loop (HIL) 3. Battery diagnostics (EIS, incremental capacity) 4. Battery ageing and degradation modelling 5. EV powertrain validation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	1) Develop and validate a modular battery digital twin library that runs in real time on HIL targets and supports both pack-level and cell-level representations, leveraging TNO's modelling portfolio and Battery Lab methodology [1][6]. 2) Develop battery diagnostics and monitoring algorithms (active and passive approaches) and integrate them into a HIL test workflow, building on published comparative analyses and impedance-related approaches [7]. 3) Demonstrate an end-to-end EV validation workflow: requirements, scenario definition, fault injection, HIL execution, and lab validation using representative battery hardware and test facilities [1][3][4]. 1. 4) Provide a pathway for generating standardised performance indicators (e.g., round-trip energy efficiency and efficiency-fade) that can support battery performance reporting and future battery passport readiness [8].			
R&D Target Specifications	Target (project end) quantitative specifications: • Real-time execution: digital twin runs at 1 ms fixed-step (or faster) on a HIL target, with numerical stability over representative WLTP and heavy-duty drive cycles. • Voltage prediction accuracy: RMS error ≤ 20 mV at cell level and ≤ 1.0% at pack voltage under dynamic current profiles. • Thermal prediction accuracy: ≤ 2 °C RMS error for key pack temperature points under combined cycling and environmental tests. • SoC estimation accuracy: ≤ 2% absolute error (0-100% SoC) during dynamic drive cycles. • SoH indicators: capacity estimation error ≤ 3% and internal resistance estimation error ≤ 5% on aged cells, using combined test data and monitoring algorithms. • Fault detection: detect selected fault classes (sensor drift, contactor anomalies, abnormal impedance growth, cooling degradation) with a true-positive rate ≥ 95% and false-alarm rate ≤ 2% under defined test scenarios. Validation environment (planned): use of high-power and climatic test capabilities as reported by TNO (e.g., 750 V / 400 A / 300 kW pack cycling and broad temperature ranges) [3][4].			
R&D Plan	Indicative 36-month R&D plan (can be aligned to the 2026 TIPA programme schedule):			



WP1 (M1-M4) Requirements and architecture

- Define use cases, HIL interfaces, data formats, and target battery system.
- Define validation scenarios and acceptance metrics.

WP2 (M2-M12) Battery model development and calibration

- Implement ECM baseline model and electro-thermal extensions.
- Parameterise using controlled pulse tests, drive-cycle tests, and impedance measurements, consistent with Battery Lab capabilities [1].
- Establish ageing model and update strategy.

WP3 (M6-M18) Diagnostics and monitoring algorithms

- Implement and benchmark passive monitoring versus active excitation based approaches, as discussed in published comparative work [7].
- Develop hybrid estimators (e.g., Kalman-filter families plus data-driven residual models).

WP4 (M10-M26) HIL integration and automated validation

- Integrate digital twin and diagnostics into the SME HIL environment.
- Develop scenario library including fault injection and fast-charging profiles.
- Continuous integration pipeline for regression testing.

WP5 (M18-M32) Lab validation and demonstrator build

- Validate models and algorithms using TNO Battery Lab and associated test facilities [1][3][4].
- Demonstrate on a representative battery module or pack and a controller stack.

WP6 (M30-M36) Industrialisation and dissemination

- Prepare documentation, APIs, and licensing model.
- Joint publications and roadmap toward commercial deployment.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**TNO (research institute) expected roles:** • Lead battery model and algorithm development (ECM, electro-thermal, ageing, diagnostics) in alignment with TNO Battery Lab research and tooling [1][6]. • Provide test design, measurement protocols, and data analysis methods. • Execute validation experiments using TNO facilities (Battery Lab and ICSP where relevant) [1][2][3][4]. • Deliver reference implementations and documentation for integration in HIL environments. • Lead publication and IP strategy, including joint dissemination with the Korean SME partner.
Expected R&R of Korean SME Partner	**Korean SME partner expected roles:** • Provide industrial HIL platforms, controller integration, and EV validation workflows. • Implement real-time deployment on target hardware and integrate with VCU/BMS/charger controllers. • Provide industrial requirements, representative use cases, and field data (where available). • Co-develop scenario libraries and support demonstrator integration and commercial packaging.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes: 1) A validated, modular battery digital twin and diagnostics software suite, ready for deployment in industrial HIL systems [1][6][7]. 2) Demonstrated reduction of test and iteration time for EV powertrain and charging control development, using HIL workflows and lab-validated models [1][3][4]. 3) Joint IP (software modules, calibration procedures, scenario library) and joint scientific outputs (conference and journal papers), leveraging TNO publication pathways [5][6][7]. 4) Commercialisation pathway: a productised validation toolkit for EV developers and Tier-1 suppliers, and service offerings based on TNO test facilities [1][2]. 5) Improved readiness for battery performance reporting and emerging battery passport related indicators [8].

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization